

More Complicated Logarithmic Equations (Part II)

Exponent Rules Learned in Pre-Algebra

I) $x^a \cdot x^b = x^{a+b}$ Example: $x^2 \cdot x^5 = x^{2+5} = x^7$

II) $\frac{x^a}{x^b} = x^{a-b}$, $x \neq 0$ Example: $\frac{x^9}{x^3} = x^{9-3} = x^6$

New Exponent Rules

III) $(xy)^a = x^a \cdot y^a$ Example: $(2m)^4 = 2^4 \cdot m^4$

IV) $\left(\frac{x}{y}\right)^a = \frac{x^a}{y^a}$, $y \neq 0$ Example: $\left(\frac{n}{3}\right)^7 = \frac{n^7}{3^7}$

V) $(x^a)^b = x^{ab}$ Example: $(x^{10})^3 = x^{10 \cdot 3} = x^{30}$

Strange New Exponent Rules

VI) $x^1 = x$ Example: $3^1 = 3$

VII) $x^0 = 1$, $x \neq 0$ Example: $5^0 = 1$

VIII) $x^{-a} = \frac{1}{x^a}$, $x \neq 0$ Example: $x^{-2} = \frac{1}{x^2}$

IX) $x^{\frac{1}{a}} = \sqrt[a]{x}$ Example: $x^{\frac{1}{3}} = \sqrt[3]{x}$

X) $x^{\frac{b}{a}} = (\sqrt[a]{x})^b$ or $\sqrt[a]{x^b}$ Example: $x^{\frac{5}{7}} = (\sqrt[7]{x})^5$ or $\sqrt[7]{x^5}$

Logarithm Property V

$$\log_k(AB) = \log_k A + \log_k B$$

Logarithm Property VI

$$\log_k\left(\frac{A}{B}\right) = \log_k A - \log_k B$$

Logarithm Property VII

$$\log_k A^B = B \log_k A$$

Solve. Write the Roman numeral of each logarithm property or exponent rule that you use.

Problems Involving Quadratic Equations (The Factoring Method)

① $\log_6 x + \log_6 (x-5) = 2$

② $3 = \log_2 (y+7) + \log_2 y$

LI $\log_6 [x(x-5)] = 2$

LII $3 = \log_2 [(y+7)y]$

$$\log_6 [x^2 - 5x] = 2$$

$$3 = \log_2 [y^2 + 7y]$$

$$6^2 = x^2 - 5x$$

$$2^3 = y^2 + 7y$$

Quadratic
Equation
Standard
Form

$$\rightarrow 36 = x^2 - 5x$$

$$\rightarrow 0 = x^2 - 5x - 36$$

$$0 = (x-9)(x+4)$$

$$\boxed{x=9} \text{ or } x \neq -4$$

$$8 = y^2 + 7y$$

$$0 = y^2 + 7y - 8$$

$$0 = (y+8)(y-1)$$

$$y \neq -8 \text{ or } \boxed{y=1}$$

$$\textcircled{3} 1 = \log_3(x+2) + \log_3 x$$

$$\text{LVI} \quad 1 = \log_3[(x+2)x] \quad \text{LVI}$$

$$1 = \log_3[x^2 + 2x]$$

$$3^1 = x^2 + 2x$$

$$\text{EVI} \quad 3 = x^2 + 2x$$

$$0 = x^2 + 2x - 3$$

$$0 = (x+3)(x-1)$$

$$\swarrow \quad \searrow$$

$$x \neq -3 \quad \text{or} \quad \boxed{x=1}$$

$$\textcircled{4} \log_4 b + \log_4(b-12) = 3$$

$$\text{LVI} \quad \log_4[b(b-12)] = 3 \quad \text{LVI}$$

$$\log_4[b^2 - 12b] = 3$$

$$4^3 = b^2 - 12b$$

$$64 = b^2 - 12b$$

$$0 = b^2 - 12b - 64$$

$$0 = (b-16)(b+4)$$

$$\swarrow \quad \searrow$$

$$\boxed{b=16} \quad \text{or} \quad b \neq -4$$

$$\textcircled{5} \log_6(x-2) + \log_6(x+3) = 2$$

$$\text{LVI} \quad \log_6[(x-2)(x+3)] = 2 \quad \text{LVI}$$

FOIL

$$\log_6[x^2 + 3x - 2x - 6] = 2$$

$$\log_6[x^2 + x - 6] = 2$$

$$6^2 = x^2 + x - 6$$

$$36 = x^2 + x - 6$$

$$0 = x^2 + x - 42 \quad \text{EVI}$$

$$0 = (x+7)(x-6)$$

$$\swarrow \quad \searrow$$

$$x \neq -7 \quad \text{or} \quad \boxed{x=6}$$

$$\textcircled{6} 1 = \log_3(x-4) + \log_3(x-6)$$

$$\text{LVI} \quad 1 = \log_3[(x-4)(x-6)] \quad \text{LVI}$$

$$1 = \log_3[x^2 - 6x - 4x + 24]$$

$$1 = \log_3[x^2 - 10x + 24]$$

$$3^1 = x^2 - 10x + 24$$

$$3 = x^2 - 10x + 24$$

$$0 = x^2 - 10x + 21$$

$$0 = (x-7)(x-3)$$

$$\swarrow \quad \searrow$$

$$\boxed{x=7} \quad \text{or} \quad x \neq 3$$

$$\textcircled{7} 1 = \log_7(x^2 + 10x + 23)$$

$$7^1 = x^2 + 10x + 23$$

EVI

$$7 = x^2 + 10x + 23$$

$$0 = x^2 + 10x + 16$$

$$0 = (x+8)(x+2)$$

$$\boxed{x = -8} \text{ or } \boxed{x = -2}$$

EVI

$$\textcircled{8} \log_2(x^2 - 3x + 4) = 1$$

$$2^1 = x^2 - 3x + 4$$

$$2 = x^2 - 3x + 4$$

$$0 = x^2 - 3x + 2$$

$$0 = (x-1)(x-2)$$

$$\boxed{x = 1} \text{ or } \boxed{x = 2}$$

Solving by Factoring Expressions of the Form $ax^2 + bx + c$

$$\textcircled{9} \log_4(2x^2 - 13x + 36) = 2$$

$$4^2 = 2x^2 - 13x + 36 \quad \text{LVI}$$

$$16 = 2x^2 - 13x + 36$$

$$0 = 2x^2 - 13x + 20$$

$$0 = (2x-5)(x-4)$$

$$2x-5=0 \text{ or } x=4$$

$$\boxed{x = \frac{5}{2}} \text{ or } \boxed{x = 4}$$

$$\textcircled{10} 2 = \log_5(x + \frac{25}{6}) + \log_5(6x)$$

$$2 = \log_5[(x + \frac{25}{6})(6x)]$$

$$2 = \log_5[6x^2 + 25x]$$

$$5^2 = 6x^2 + 25x$$

$$25 = 6x^2 + 25x$$

$$0 = 6x^2 + 25x - 25$$

$$0 = (6x-5)(x+5)$$

$$6x-5=0 \text{ or } x=-5$$

$$\boxed{x = \frac{5}{6}} \text{ or } x \neq -5$$

$$\textcircled{11} 0 = \log_3(5x^2 + 6x + 2)$$

$$3^0 = 5x^2 + 6x + 2$$

LVI

$$\textcircled{12} \log_2(3x) + \log_2(x - \frac{1}{3}) = 1$$

$$\log_2[3x(x - \frac{1}{3})] = 1$$

$$\log_2[3x^2 - x] = 1$$

$$2^1 = 3x^2 - x$$

$$2 = 3x^2 - x$$

$$0 = 3x^2 - x - 2$$

$$0 = (3x + 2)(x - 1)$$

$$3x + 2 = 0 \text{ or } x = 1$$

$$x \neq -\frac{2}{3} \text{ or } \boxed{x = 1}$$

EVI

EVII

$$1 = 5x^2 + 6x + 2$$

$$0 = 5x^2 + 6x + 1$$

$$0 = (5x + 1)(x + 1)$$

$$\swarrow \quad \searrow \quad x = -1$$

$$5x + 1 = 0$$

$$\boxed{x = -\frac{1}{5}} \text{ or } \boxed{x = -1}$$

Equations Requiring the Quadratic Formula

$$\textcircled{13} \log_5 x + \log_5(x - 2) = 1$$

LVI

$$\log_5[x(x - 2)] = 1$$

$$\log_5[x^2 - 2x] = 1$$

$$5^1 = x^2 - 2x$$

$$5 = x^2 - 2x$$

$$0 = x^2 - 2x - 5$$

$$a = 1, b = -2, c = -5$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$x = \frac{-(-2) \pm \sqrt{(-2)^2 - 4(1)(-5)}}{2(1)}$$

$$x = \frac{2 \pm \sqrt{4 + 20}}{2}$$

$$x = \frac{2 \pm \sqrt{24}}{2}$$

$$x = \frac{2 \pm 2\sqrt{6}}{2}$$

$$x = 1 \pm \sqrt{6}$$

$$\boxed{x = 1 + \sqrt{6}} \text{ or } \boxed{x = 1 - \sqrt{6}}$$

$$\textcircled{14} 0 = \log_4(x+2) + \log_4(3x)$$

$$\text{LII} \quad 0 = \log_4[(x+2)(3x)]$$

$$0 = \log_4[3x^2 + 6x]$$

$$4^0 = 3x^2 + 6x$$

$$\text{EIII} \quad 1 = 3x^2 + 6x$$

$$0 = 3x^2 + 6x - 1$$

$$a=3, b=6, c=-1$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$x = \frac{-6 \pm \sqrt{6^2 - 4(3)(-1)}}{2(3)}$$

$$x = \frac{-6 \pm \sqrt{36 + 12}}{6}$$

$$x = \frac{-6 \pm \sqrt{48}}{6}$$

$$x = \frac{-3 + 2\sqrt{3}}{3} \text{ or } x \neq \frac{-3 - 2\sqrt{3}}{3}$$

$$x = \frac{-6 \pm 4\sqrt{3}}{6}$$

$$x = \frac{-3 \pm 2\sqrt{3}}{3}$$

$$\textcircled{15} \log_2(x+5) + \log_2 x = 2$$

$$\text{LVI} \quad \log_2[(x+5)x] = 2$$

$$\log_2[x^2 + 5x] = 2$$

$$2^2 = x^2 + 5x$$

$$4 = x^2 + 5x$$

$$0 = x^2 + 5x - 4$$

$$a=1, b=5, c=-4$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$x = \frac{-5 \pm \sqrt{5^2 - 4(1)(-4)}}{2(1)}$$

$$x = \frac{-5 \pm \sqrt{25 + 16}}{2}$$

$$x = \frac{-5 \pm \sqrt{41}}{2}$$

$$x = \frac{-5 + \sqrt{41}}{2} \text{ or } x \neq \frac{-5 - \sqrt{41}}{2}$$

$$(16) 0 = \log_3(4x) + \log_3(x-3)$$

$$\text{LVI} \quad 0 = \log_3[4x(x-3)]$$

$$0 = \log_3[4x^2 - 12x]$$

$$3^0 = 4x^2 - 12x$$

$$\text{EVII} \quad 1 = 4x^2 - 12x$$

$$0 = 4x^2 - 12x - 1$$

$$a=4, b=-12, c=-1$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$x = \frac{-(-12) \pm \sqrt{(-12)^2 - 4(4)(-1)}}{2(4)}$$

$$x = \frac{12 \pm \sqrt{144 + 16}}{8}$$

$$x = \frac{12 \pm \sqrt{160}}{8}$$

$$x = \frac{12 \pm 4\sqrt{10}}{8}$$

$$x = \frac{3 \pm \sqrt{10}}{2}$$

$$x = \frac{3 + \sqrt{10}}{2} \text{ or } x \neq \frac{3 - \sqrt{10}}{2}$$

More Problems Involving Quadratic Equations

$$(17) \log_{(x+1)}(-2x+46) = 2$$

$$(x+1)^2 = -2x+46$$

$$(x+1)(x+1) = -2x+46$$

FOIL

$$x^2 + x + x + 1 = -2x + 46$$

$$x^2 + 2x + 1 = -2x + 46$$

$$x^2 + 4x - 45 = 0$$

$$(x+9)(x-5) = 0$$

$$x \neq -9 \text{ or } x = 5$$

EVII

$$\star (19) \log_{(x-5)}(2x^2 - 3x + 17) = 2$$

$$(x-5)^2 = 2x^2 - 3x + 17$$

$$(x-5)(x-5) = 2x^2 - 3x + 17$$

FOIL

$$x^2 - 5x - 5x + 25 = 2x^2 - 3x + 17$$

$$x^2 - 10x + 25 = 2x^2 - 3x + 17$$

$$0 = x^2 + 7x - 8$$

$$0 = (x+8)(x-1)$$

$$x \neq -8 \text{ or } x \neq 1$$

No Solution

EVII

$$(18) 1 = \log_{(x-3)}(33 - 11x + x^2)$$

$$(x-3)^1 = 33 - 11x + x^2$$

$$x-3 = 33 - 11x + x^2$$

$$0 = 36 - 12x + x^2$$

$$0 = x^2 - 12x + 36$$

$$0 = (x-6)(x-6)$$

$$x = 6$$

$$(20) 1 = \log_{(4x-9)}(x^2 + 2x - 12)$$

$$(4x-9)^1 = x^2 + 2x - 12$$

$$4x-9 = x^2 + 2x - 12$$

$$0 = x^2 - 2x - 3$$

$$0 = (x-3)(x+1)$$

$$x = 3 \text{ or } x \neq -1$$